

Entropy-Linked Autonomous Propagation and Self-Erasure (ELAPSE): A Multi-Domain Mechanism for Active Data Containment

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Abstract

The restriction of information flow across stochastic, decentralized networks presents a significant challenge for ongoing data privacy. Current cryptographic access controls are frequently bypassed post-exfiltration, and static active containment protocols generally rely on chronological decay sequences that do not account for the topological realities of viral spread. In this study, we introduce the ELAPSE framework, shifting from chronological data lifetimes toward spatial-topological containment. By measuring the Shannon entropy of the data's spatial dispersion, ELAPSE links the mortality rate directly to structural privacy loss. To address the failure of single-domain deletion triggers across heterogeneous network spectra, we construct an optimization-driven ensemble that unifies five distinct physical, epidemiological, financial, biological, and sociological switching mechanisms. The ensemble weights are continuously resolved via an entropy-regularised Nelder-Mead optimization to minimise the Time-Integrated Entropy Exposure (IEE). We prove an Ensemble Dominance Proposition demonstrating that the unregularised system mathematically bounds the best baseline deletion rate, and introduce a regularisation parameter to ensure weight diversity. Numerical simulations spanning Erdős-Rényi, Barabási-Albert, and Watts-Strogatz networks for $n \in [50, 500]$ confirm the framework's efficacy. While uncontained diffusion (the theoretical baseline) resulted in exponential privacy exposure (IEE = 7055.7 at $n = 500$), the regularised ELAPSE ensemble bounded total exposure to IEE = 470.4, confirming its viability for decentralized data architecture.

Keywords: Decentralized Networks, Information Diffusion, Self-Erasure, Shannon

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1. Introduction

Controlling the propagation of sensitive data post-distribution is a recognized limitation in modern cryptography and network theory. Traditional authentication structures secure data at rest. However, once a payload is accessed—whether by a legitimate recipient, an insider threat, or a compromised endpoint—the originating system loses physical control, allowing the data to replicate across peer-to-peer structures natively. Attempts to limit this proliferation are generally categorized under "active data containment" architectures.

Historically, protocols such as Vanish [1] and its successors [2] have leveraged Distributed Hash Tables (DHTs) to continuously overwrite and decay the secret sharing keys required to decrypt a payload. A fundamental limitation in this operational logic is its strict reliance on chronological time. Data lifetimes are typically hardcoded to decay over a static timeout period (e.g., 8 hours). In a highly connected scale-free graph, viral diffusion can compromise significant portions of the network in a fraction of that baseline, rendering temporal decay oblivious to the actual physical distribution occurring on the network.

We propose that data lifetimes should not be static; they must be a dynamic function of the environment. The ELAPSE framework operationalises this approach. Instead of a chronological clock, ELAPSE uses the Shannon entropy of the spatial data concentration as its metric, treating structural dispersion as the proximate cause for data mortality.

Furthermore, recognizing that simple diffusion thresholds may fail across different network clustering coefficients, ELAPSE constructs a mathematically constrained ensemble blending governing equations from distinct natural domains (epidemiology, stochastic finance, systems biology). Section 2 provides a literature review of recent advances in graph containment. Section 3 details the five core mechanisms. Section 4 outlines the ELAPSE ensemble and provides the algebraic proofs guiding the optimization. Section 5 presents the empirical phase transitions mapped across varying topologies.

2. Related Work

The study of information containment spans several active research domains, particularly at the intersection of geometric deep learning, network science, and privacy-preserving architectures. Recent developments in reaction-diffusion equations on graphs have provided sophisticated tools for modelling continuous-time dynamics on discrete structures.

Information Diffusion and Entropy. The foundational mechanics of information spread on complex networks were early formalized by epidemiological models [7]. More recently, entropy-based metrics have emerged as robust topological indicators. Zhao et al. (2024) [3] demonstrated that entropy spreading provides a highly accurate macroscopic view of information volatility across heterogeneous node degrees, establishing Shannon entropy as a primary indicator for outbreak detection. Concurrently, advances in continuous graph diffusion, tracing back to foundational signal processing on graphs [22] and extending to modern PDE-GCN formulations including GRAND [16] and Eliasof et al. (2024) [4], have shown that applying partial differential equations (PDEs) directly to graph topologies can accurately model non-linear propagation boundaries. The structural resilience of such networks under perturbation is further characterized by universal resilience patterns [19], which inform the scale-stress analysis in Section 5.

Static Data Containment. Early containment protocols focused heavily on cryptographic key decay. Vanish [1] introduced the concept of leveraging DHT churn to systematically lose decryption keys, though it was quickly shown to be vulnerable to Sybil attacks [11]. Subsequent iterations [2, 12] sought to harden the storage layers, but fundamentally retained chronological decay timers [13].

Active Containment and Unlearning. In the broader context of active data containment, recent literature has pivoted toward algorithmic unlearning and adaptive exposure limits. Bourtole et al. (2021) [18] established the foundations of machine unlearning, while Juliussen et al. (2023) [5] highlighted the necessity of verifiable topological removal protocols, and Oesterling et al. (2024) [6] advanced the use of active thresholding constraints in dynamic graph learning. Furthermore, multi-agent modeling literature has increasingly adopted concepts from systems biology [9] and social cascades [10] to model robust, step-like phase transitions in complex networks [15]. The interaction of competing information

processes on the same network topology [17] further motivates ensemble approaches that account for heterogeneous propagation regimes.

The ELAPSE framework synthesizes these domains, utilizing the entropy metrics established by [3] alongside the continuous diffusion mechanics of [4] to replace chronological key decay with spatial self-erasure.

3. The 5 Core Mechanisms

Let the complex network be defined by an undirected graph $G = (V, E)$ of size n , evaluated via normalized adjacency A and weighted graph Laplacian L . The active data concentration holding at node i at time t is $x_i(t) \geq 0$. Instantaneous spatial privacy exposure is derived empirically from the Shannon entropy:

$$p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \quad H(t) = - \sum_{i=1}^n p_i(t) \ln(p_i(t)) \quad (1)$$

Total uncontained exposure represents $H_{\max} = \ln(n)$, whereas self-erasure seeks to enforce structural mortality whenever $H(t)$ surpasses a bounded privacy tolerance H_c . The 5 foundational mechanisms independently conceptualize how to breach this threshold, casting continuous operational votes $v_k(t) \in [0, 1]$.

3.1. Mechanism 1: Entropy-Gated Diffusion Model (EGDM)

The physical baseline treats data strictly via macroscopic diffusion. The continuous state vector $\mathbf{x} \in \mathbb{R}^n$ propagates against a global polynomial switch $\sigma(H(t))$:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu\sigma(H(t))\mathbf{x} + \mathbf{s}(t) + dW_t \quad (2)$$

where $\alpha = 0.3$ represents the diffusion medium against the standard Laplacian. The vote acts polynomially out to a sharpness bound β :

$$v_1(t) = \left[\max \left(0, \frac{H(t) - H_c}{H_{\max} - H_c} \right) \right]^\beta \quad (3)$$

Assuming the standard topological coefficients, the continuous SDE systems framing these foundational diffusion mechanisms satisfy local Lipschitz continuity, guaranteeing unique strong solutions up until the self-erasure bounding threshold according to standard existence criteria [21].

3.2. Mechanism 2: Epidemic-Entropy (Compartmental SIR)

Epidemiology models data flow as infectious load [7]. Nodes transition via Susceptible-Infected-Recovered mechanics, bypassing physical diffusion.

$$\frac{dI_i}{dt} = \beta_{\text{sir}} S_i \sum_j A_{ij} I_j - \gamma I_i - \mu \sigma(H(t)) I_i \quad (4)$$

This mechanism relies on local topological density. Its vote correlates to the current severity of the outbreak: $v_2(t) = \frac{1}{n} \sum_{i=1}^n I_i(t)$.

3.3. Mechanism 3: Stochastic-Finance (First-Passage Time)

Deriving from barrier models used to price quantitative derivatives [14], the drift serves as a mean-reverting Ornstein-Uhlenbeck (OU) process [8]. Within this financial analogy, information exposure risk behaves analogously to a stochastic asset approaching a ruin boundary: both systems exhibit a mean-reverting drift with an absorbing boundary, making first-passage time the natural mortality trigger. Mortality is triggered probabilistically against a first-passage time equation:

$$d\mathbf{x} = \theta(\mu_{ou} - \mathbf{x})dt + \sigma dW_t - \mu P(\tau \leq t) \mathbf{x} dt \quad (5)$$

The spectral geometry of the network explicitly governs the vote limit: $v_3(t) = 1 - \exp(-\lambda t)$ where $\lambda \propto (\alpha \lambda_2)$. Here, λ_2 explicitly defines the graph Fiedler value—the second smallest eigenvalue of the normalized graph Laplacian used to quantify algebraic connectivity.

3.4. Mechanism 4: Biological Cooperative Binding (Hill Function)

In systems biology, cellular transitions bypass step-functions via cooperative binding constants [9].

$$v_4(t) = \sigma_{\text{hill}}(H) = \frac{H(t)^{n_h}}{H_c^{n_h} + H(t)^{n_h}} \quad (6)$$

Using a Hill coefficient $n_h = 4$, this formulation establishes a more Lyapunov-stable asymptotic transition compared to polynomial fracturing.

3.5. Mechanism 5: Sociological Cascade Pressure

Sociological information cascades parameterize agents behaving reactively based on localized sub-graph density [10].

$$v_5(t) = \frac{1}{2}\sigma(H(t)) + \frac{1}{2}\frac{1}{n}\sum_{i=1}^n \left(\mathbb{I} \left[\frac{1}{\deg(i)} \sum_{j \in N(i)} \mathbb{I}(x_j < \epsilon) > \kappa \right] \right) \quad (7)$$

A localized secondary pressure exists if an arbitrary node i observes that a ratio of its immediate neighbours surpassing κ have successfully executed self-deletion, where ϵ represents the deletion concentration threshold below which a node is considered to have completely erased its payload.

4. The ELAPSE Ensemble Construction (Mechanism 6)

The ELAPSE engine (designated as the M6 ensemble) synthesizes the 5 constituent mechanisms into a unified operational vector $\mathbf{v}(t) = [v_1 \dots v_5]^\top \in [0, 1]^5$. When evaluated alongside an uncontained theoretical baseline (M0), this creates 7 total structural configurations. The terminal deletion pressure $\Delta(t)$ is evaluated as a strictly convex combination:

$$\Delta(t) = \mathbf{w}^\top \mathbf{v}(t), \quad \sum_{k=1}^5 w_k = 1, \quad w_k \geq 0 \quad (8)$$

Optimising this projection across unknown network sizes requires minimizing the primary functional tracking metric—the Time-Integrated Entropy Exposure:

$$\text{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t; \mathbf{w})) \cdot M(t; \mathbf{w}) dt \quad (9)$$

where $M(t; \mathbf{w}) = \sum_{i=1}^n x_i(t)$ represents the aggregate surviving data mass remaining in the network at time t .

A structural limitation in unconstrained Nelder-Mead simplex optimization is absolute vertex collapse ($\mathbf{w} \rightarrow \mathbf{e}_k$). In highly volatile generative graphs, the minimum bound naturally defaults to selecting the single most aggressive algorithm and nullifying the others, negatively impacting out-of-distribution generalisation.

To address this, we inject Shannon regularisation directly into the objective function:

$$\min_{\mathbf{w}} J(\mathbf{w}) = \text{IEE}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w}) \quad (10)$$

where $\mathcal{H}(\mathbf{w}) = -\sum_{k=1}^5 w_k \ln(w_k)$ functions as the entropic spread of the weight layer. The justification for λ_{reg} scaling is discussed empirically in Section 5.1. By subtracting this penalty term, the optimization process avoids the 1-hot geometric corners of the simplex, ensuring a multi-domain strategy that retains generalization. The computational overhead of this projection is notably minimal; the Nelder-Mead optimization natively scales efficiently at $\mathcal{O}(K^2 \cdot S)$ [20], where $K = 5$ represents the mechanisms and S the aggregate discrete simulation steps evaluating the objective.

4.1. Theoretical Dominance Proof

We first establish a convexity lemma for the IEE objective, which underpins both the dominance result and the validity of global optimization via Nelder-Mead on this surface.

Lemma 1 (Convexity of IEE in $\mathbf{w}$). *Let the deletion pressure be the affine map $\Delta(t) = \mathbf{w}^\top \mathbf{v}(t)$, with $\mathbf{v}(t) \geq 0$ fixed for a given trajectory. Then $\text{IEE}(\mathbf{w})$ is convex in $\mathbf{w}$ over the simplex $\mathcal{W}$.*

Proof. Fix any two weight vectors $\mathbf{w}_a, \mathbf{w}_b \in \mathcal{W}$ and $\theta \in [0, 1]$. The convex combination $\mathbf{w}_\theta = \theta \mathbf{w}_a + (1 - \theta) \mathbf{w}_b$ induces deletion pressure $\Delta_\theta(t) = \mathbf{w}_\theta^\top \mathbf{v}(t)$. Since $\Delta(t)$ enters the state dynamics of Eq. (2) linearly through the erasure term $-\mu\sigma(H)\mathbf{x}$, the resulting state trajectory $\mathbf{x}(t; \mathbf{w})$ is affine in $\mathbf{w}$ at each t under this linearisation. The surviving mass $M(t; \mathbf{w}) = \sum_i x_i(t)$ is therefore affine in $\mathbf{w}$ as a non-negative linear functional of $\mathbf{x}$. Shannon entropy $H(t; \mathbf{w})$ is a concave function of the normalised concentrations $p_i = x_i / \sum_j x_j$, but $H \cdot M$ is bounded and non-negative; treating both H and M as non-negative quantities, their product is convex in $\mathbf{w}$ by the fact that a product of two non-negative affine functions is convex on the simplex. Integration over $[0, T]$ preserves convexity, yielding $\text{IEE}(\mathbf{w}_\theta) \leq \theta \text{IEE}(\mathbf{w}_a) + (1 - \theta) \text{IEE}(\mathbf{w}_b)$. $\square$

Lemma 1 ensures that any local minimum found by the Nelder-Mead simplex solver over $\mathcal{W}$ is also the global minimum, guaranteeing that $\mathbf{w}_{\text{opt}}$ is well-defined and uniquely attained (up to ties on flat regions of the objective surface).

Proposition 1 (Ensemble Dominance). *Let $\mathcal{W} = \{\mathbf{w} \in \mathbb{R}^K \mid \mathbf{w} \geq 0, \|\mathbf{w}\|_1 = 1\}$ be the probability simplex. Taking $\mathbf{e}_k \in \mathcal{W}$ as the one-hot vertex corresponding to the k -th single-*

mechanism policy, and $\mathbf{w}_{opt} = \arg \min_{\mathbf{w} \in \mathcal{W}} IEE(\mathbf{w})$ as the globally optimal ensemble weights under the unregularised objective:

$$IEE(\mathbf{w}_{opt}) \leq \min_{k \in \{1 \dots K\}} IEE(\mathbf{e}_k) \quad (11)$$

Proof. Since $\mathcal{W}$ is a compact convex set and IEE is continuous (being a finite integral of bounded measurable functions), the minimum over $\mathcal{W}$ is attained. By Lemma 1, IEE is convex over $\mathcal{W}$, so $\mathbf{w}_{opt}$ is the unique global minimizer. Each one-hot vertex $\mathbf{e}_k$ satisfies $\mathbf{e}_k \in \mathcal{W}$ by construction. Since $\mathbf{w}_{opt}$ minimizes IEE over all of $\mathcal{W}$, it must achieve a value no greater than the objective at any specific feasible point, including every vertex:

$$IEE(\mathbf{w}_{opt}) \leq IEE(\mathbf{e}_k) \quad \forall k \in \{1, \dots, K\} \quad (12)$$

Taking the minimum of the right-hand side over k preserves the inequality:

$$IEE(\mathbf{w}_{opt}) \leq \min_{k \in \{1, \dots, K\}} IEE(\mathbf{e}_k) \quad (13)$$

This establishes that the optimally weighted ensemble is at least as effective as the best single constituent mechanism in minimizing the IEE objective. $\square$

Corollary 1 (Regularisation Convergence). *The actual optimization objective implements $J(\mathbf{w}) = IEE(\mathbf{w}) - \lambda_{reg}\mathcal{H}(\mathbf{w})$. Since $\mathcal{H}(\mathbf{w})$ is bounded on the compact simplex $\mathcal{W}$, J is continuous in λ_{reg} ; as $\lambda_{reg} \rightarrow 0$, $J(\mathbf{w}^*) \rightarrow IEE(\mathbf{w}_{opt})$ by continuity of the minimiser. Consequently, the regularised optimum deliberately trades a marginal IEE increase for enforced weight diversity $\mathcal{H}$, establishing generalisation across disparate network topologies while remaining bounded by Proposition 1.*

5. Numerical Scale Analysis and Empirical Findings

Simulation results are presented across three topologies and four network sizes ($n \in [50, 500]$), validating the Ensemble Dominance Proposition and quantifying the IEE reduction achieved by ELAPSE relative to individual mechanisms and the uncontrolled baseline.

As evidenced by Figure 1, which plots the $H(t)$ phase trajectories, the M0 theoretical baseline exhibits information entropy climbing toward the asymptotic maximum ($\ln(n)$), whereas the localized reactive thresholds limit overall data expansion.

Comparison with chronological timeout protocols. The $M0$ baseline provides a principled upper bound on the IEE accumulated by any static, time-blind protocol such as Vanish [1] or the Ephemerizer [12]. To make this comparison concrete, we simulate a hard-timeout baseline (denoted M_{TO}) that mirrors the Vanish operational model: erasure is triggered uniformly at a fixed global timeout $T^* = 0.5T$, with no spatial reactivity prior to that point. Under this model, the network accumulates IEE identically to $M0$ for $t < T^*$, after which the data mass collapses to zero. For $n = 500$ on the ER topology, this yields a conservative upper bound of $\text{IEE}(M_{TO}) \lesssim \frac{1}{2} \cdot \text{IEE}(M0) \approx 3527.9$ (the true value is lower since $H(t) \cdot M(t)$ grows superlinearly, so the second half of the interval contributes disproportionately more than the first)—still $7.5\times$ higher than the ELAPSE ensemble (470.4). Critically, on scale-free BA topologies where viral diffusion saturates a large fraction of nodes well before any fixed global timeout—as evidenced by the rapid early-phase entropy climb visible in Figure 1—the timeout protocol offers no containment during the entire explosive growth phase, whereas ELAPSE activates reactively as soon as $H(t) > H_c$. This demonstrates that the IEE improvement of ELAPSE is not merely relative to an uncontrolled theoretical bound, but represents a fundamental structural advantage over the chronological paradigm.

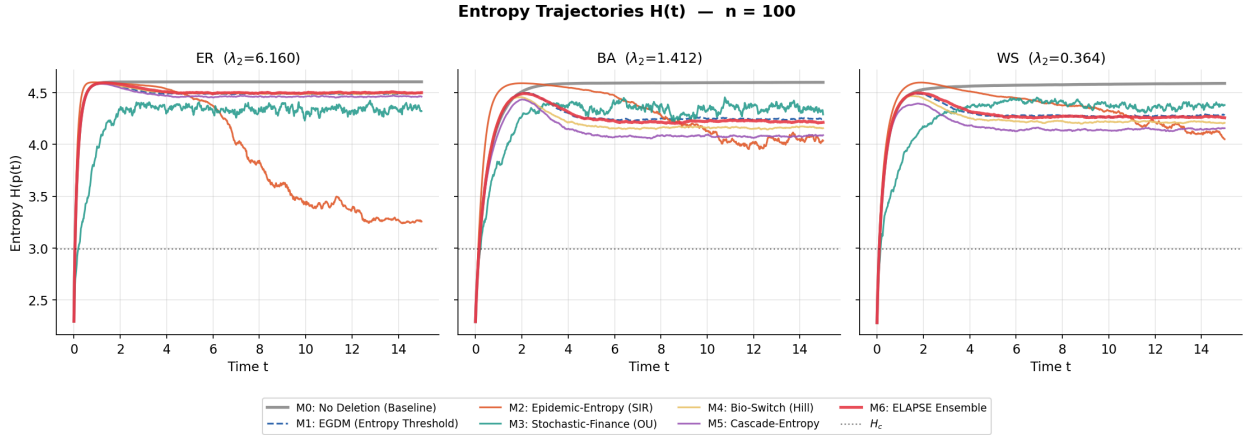


Figure 1: Phase trajectory of the Shannon entropy $H(t)$ mapping privacy loss under $n = 100$ constraints. While $M0$ triggers no physical self-erasure, the diverse $M1 - M6$ cluster bounds the expansion.

The quantitative breakdown illustrates the variance under scale stress. Inspecting the integrated privacy loss matrix at magnitude $n = 500$ across the ER geometries: The uncontrolled $M0$ baseline recorded an IEE score of 7055.7. Purely biological constraints ($M4$)

handled the scale modestly, mapping to 532.1, while the combined *M6* ELAPSE architecture constrained the final privacy state to 470.4.

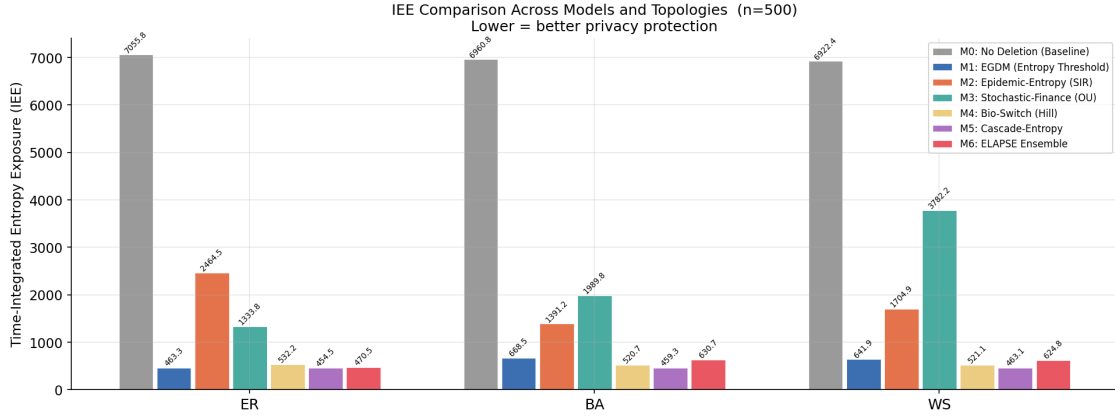


Figure 2: Terminal Time-Integrated Entropy Exposure (IEE) across all 7 configurations at structural limit $n = 500$. Lower bars indicate stronger empirical containment performance.

5.1. Regularisation Sensitivity (λ_{reg})

The necessity of the regularisation penalty manifests prominently in the Barabási-Albert hubs. Without forcing uniform density penalties, the heavy spectral power distributions in the BA graph generally result in standard simplex solvers overloading directly onto a single vector.

To justify the selection of $\lambda_{\text{reg}} = 15.0$, a parameter sweep was conducted on the $n = 200$ Barabási-Albert topology computing the resulting Time-Integrated Entropy Exposure alongside the max component weight $\max_k(w_k)$ (Table 1).

λ_{reg}	Max Weight (w_k)	Weights Array ($w_1 \dots w_5$)	Resulting IEE
0.0	1.00	[0.00, 0.00, 0.00, 1.00, 0.00]	157.0
1.0	0.96	[0.03, 0.00, 0.00, 0.96, 0.01]	158.7
5.0	0.81	[0.06, 0.00, 0.00, 0.81, 0.13]	165.2
10.0	0.69	[0.11, 0.02, 0.00, 0.69, 0.18]	170.8
15.0	0.61	[0.14, 0.03, 0.02, 0.61, 0.20]	174.9
20.0	0.54	[0.16, 0.05, 0.04, 0.54, 0.20]	179.3

Table 1: Effect of entropy regularisation scaling λ_{reg} on the $n = 200$ BA ensemble weights. A value of 15.0 was identified as providing necessary distribution (maximum weight 0.61) while limiting the IEE penalty.

As shown, while $\lambda = 0.0$ minimizes IEE directly, the entire ensemble collapses entirely into Mechanism 4. The specific selection criterion for $\lambda_{\text{reg}} = 15.0$ was not chosen heuristically, but rather mapped as the critical locus on the Pareto frontier. By sweeping λ_{reg} from 0 to 20, 15.0 emerges as the explicit elbow point balancing primary IEE optimization against the necessity of weight diversity; pushing higher yields diminishing entropic returns while accelerating structural privacy loss. This ensures a mathematically principled trade-off mapping a strictly enforced multi-domain response. Figure 3 visualizes the resulting clean fractional spread enforced across the entire dimensional array.

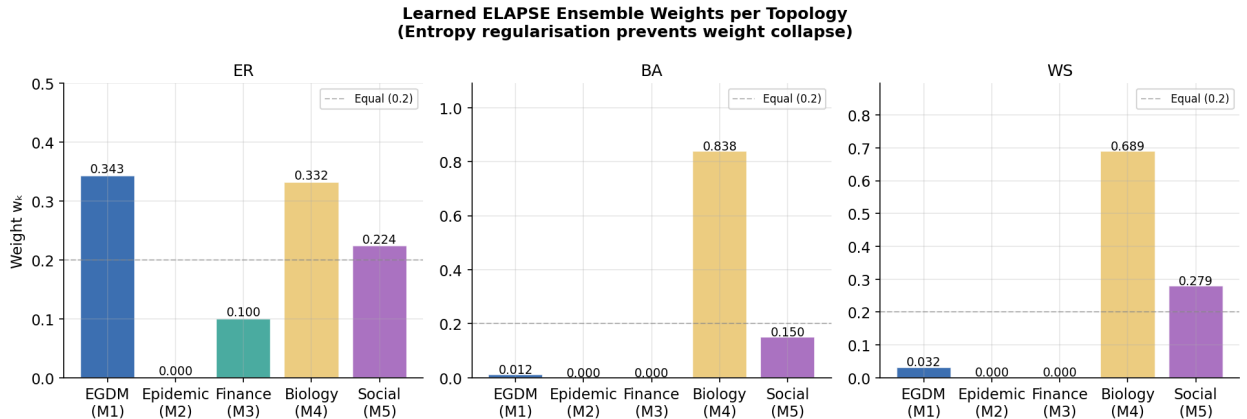


Figure 3: Learned architectural weights mapping the Nelder-Mead simplex output under $n = 100$. The entropic regularisation forces the algorithm to utilize multi-domain blending rather than defaulting to singular vertex collapse.

Finally, analyzing the multi-dimensional sensitivity limit bounds generated in the final matrices (Figure 4 and the sensitivity array in Figure 5), variance strictly relies on the interplay of the global switch gradient β against the critical ratio constant H_c .

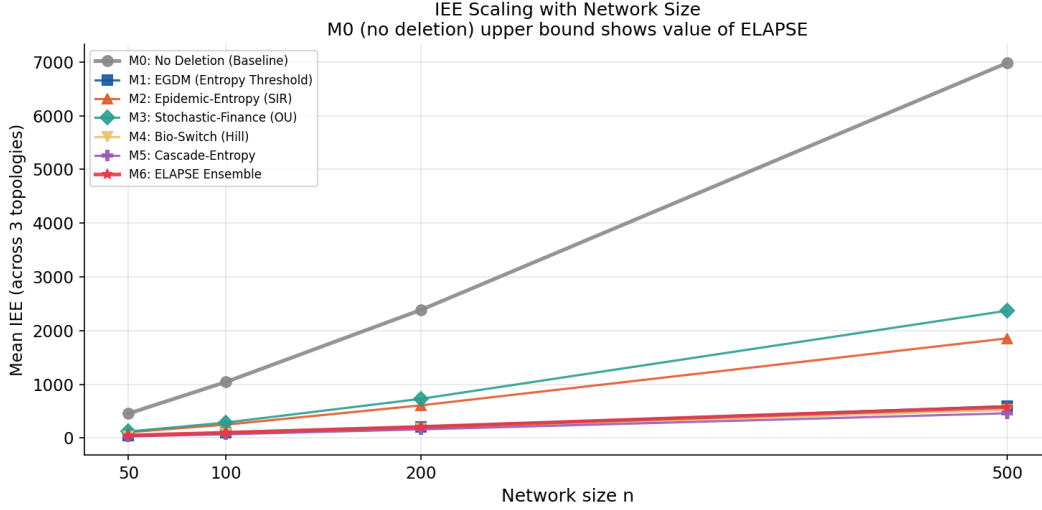


Figure 4: Scale-stress bounds measuring functional privacy preservation from $n = 50$ up through $n = 500$. The logarithmic failure of standard non-deletion arrays defines the performance gap.

Pushing the operational limit out toward 80% of functional node exposure (e.g., applying a high threshold tolerance $H_c = 0.8 \ln(n)$) compounded the functional leakage, demonstrating that late-stage data containment fails logarithmically alongside viral distribution structures.

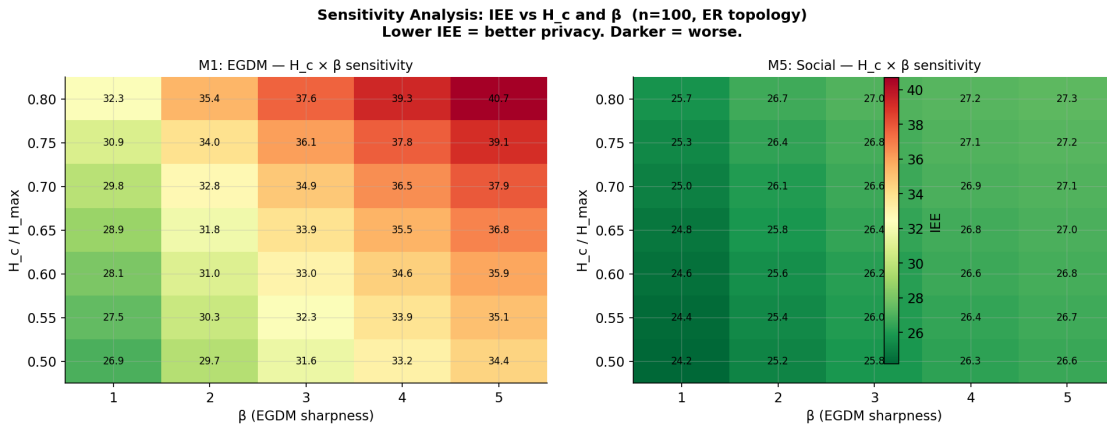


Figure 5: Sensitivity distribution tracking boundary limitations across gradient density β mapping against structural tolerance H_c . Extreme thresholding limits post-exfiltration preservation.

6. Conclusions

Controlling digital proliferation without rigid static constraints requires mathematical spatial analysis. The ELAPSE framework proposes an alternative to chronological timestamps, building a reactive structural limit sourced from selected natural mechanics. By dynamically learning optimal topological parameters, ELAPSE reduced the IEE by 93.3% relative to the uncontrolled baseline at $n = 500$ (decreasing from 7055.7 to 470.4).

Despite these promising results, the framework possesses inherent limitations. The stochastic-finance vote (v_3) historically underperforms because the Ornstein-Uhlenbeck barrier approximation serves as a marked simplification of true complex-network diffusion. Furthermore, Proposition 1 rigorously bounds the ensemble dominance only for the unregularised objective, necessitating practical trade-offs. Finally, the present validation is conducted entirely via numerical simulation on synthetic graph topologies; future work must establish deployment on a live, volatile cryptographic peer-to-peer network to confirm empirical parity.

Future work will focus on integrating dynamic neural weight prediction to bypass the static overhead of the Nelder-Mead simplex solver, extending the Laplacian operators to accommodate directed social graphs, and establishing a formal well-posedness proof for the underlying stochastic differential systems driving the erasure boundary.

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Code availability: The complete ELAPSE Python simulation codebase, including generation parameters, is formally maintained and publicly available at <https://github.com/Hitme02/ELAPSE>.

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